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Culture plate

Abstract

A culture plate including: an array of light sources formed inside and on top of a semiconductor substrate; a transparent planarization layer coating the array of light sources; and an array of wells formed on said layer, each light source being located vertically in line with a well.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION

(1) This application claims priority to French Patent Application 2006151, filed Jun. 12, 2020, the contents of which is incorporated herein by reference in its entirety.

TECHNICAL BACKGROUND

(2) The present disclosure generally concerns electronic devices and, more particularly, devices of stimulation of cells with light.

PRIOR ART

(3) In fields such as optogenetics, light sources associated with a microscope are often used to illuminate cultured cells. Tissues of living organisms may alternately be illuminated due to optical fibers coupled to a light source. Tools coupled with microscopy, which allow a spatial addressing of the emitted light, are further known. Current illumination devices however remain limited, particularly in terms of illuminated surface area and of illumination parameters capable of being tested in parallel.

SUMMARY

(4) There is a need to improve current cell illumination devices and methods of manufacturing the same.

(5) An embodiment overcomes all or part of the disadvantages of known cell illumination devices and methods of manufacturing the same.

(6) An embodiment provides a culture plate comprising: an array of light sources formed inside and on top of a semiconductor substrate; a transparent planarization layer coating the array of light sources; and an array of wells formed on said layer, each light source being located vertically in line with a well.

(7) According to an embodiment, said layer is made of a photocrosslinkable organosilicon material.

(8) According to an embodiment, said layer has a thickness in the range from 1 μm to 10 μm .

(9) According to an embodiment, the well array is made of polydimethylsiloxane.

(10) According to an embodiment, each light source comprises at least one organic light-emitting diode.

(11) According to an embodiment, each light source comprises a diode intended to emit white light.

(12) According to an embodiment, each diode is individually controllable.

(13) According to an embodiment, each light source comprises three diodes intended to respectively emit red light, green light, and blue light.

(14) According to an embodiment, the array of light sources is located on the side of a first surface of the substrate, opposite to a second surface of the substrate comprising connection pads.

(15) According to an embodiment, the connection pads are connected to the first surface of the substrate by through vias.

(16) According to an embodiment, each light source comprises a control circuit independent from the other light sources.

(17) According to an embodiment, the bottom and/or the lateral walls of at least a portion of the wells of the array are functionalized.

(18) An embodiment provides a method of manufacturing a plate such as described.

(19) According to an embodiment, the method comprises the steps of: forming the array of light sources inside and on top of the semiconductor substrate; coating the array of light sources with the transparent planarization layer; and forming at the surface of said layer a well vertically in line with each light source.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) The foregoing and other features and advantages of the present invention will be discussed in detail in the following non-limiting description of specific embodiments and implementation modes in connection with the accompanying drawings, in which:
- (2) FIG. 1 is a partial simplified perspective view of an embodiment of a multi-well culture plate;
- (3) FIG. 2 is a partial simplified cross-section view illustrating a step of an implementation mode of a method of manufacturing the culture plate of FIG. 1;
- (4) FIG. 3 is a partial simplified perspective view illustrating another step of the implementation mode of the method of manufacturing the culture plate of FIG. 1;
- (5) FIG. 4 is a partial simplified cross-section view illustrating still another step of the implementation mode of the method of manufacturing the culture plate of FIG. 1;
- (6) FIG. 5 is a partial simplified cross-section view illustrating still another step of the implementation mode of the method of manufacturing the culture plate of FIG. 1;
- (7) FIG. 6 is a partial simplified cross-section view illustrating still another step of the implementation mode of the method of manufacturing the culture plate of FIG. 1; and
- (8) FIG. 7 is a partial simplified cross-section view illustrating still another step of the implementation mode of the method of manufacturing the culture plate of FIG. 1.

DESCRIPTION OF THE EMBODIMENTS

(9) Like features have been designated by like references in the various figures. In particular, the structural and/or functional elements common to the different embodiments and implementation modes may be designated with the same reference numerals and may have identical structural, dimensional, and material properties.

(10) For clarity, only those steps and elements which are useful to the understanding of the described embodiments and implementation modes have been shown and will be detailed. In particular, the organic light-emitting diodes and their manufacturing methods are not detailed, the embodiments and implementation modes being compatible with current organic light-emitting diodes and methods of manufacturing the same.

(11) Unless indicated otherwise, when reference is made to two elements connected together, this signifies a direct connection without any intermediate elements other than conductors, and when reference is made to two elements coupled together, this signifies that these two elements can be connected or they can be coupled via one or more other elements.

(12) In the following description, when reference is made to terms qualifying absolute positions, such as terms “front”, “rear”, “top”, “bottom”, “left”, “right”, etc., or relative positions, such as terms “above”, “under”, “upper”, “lower”, etc., or to terms qualifying directions, such as terms “horizontal”, “vertical”, etc., it is referred unless specified otherwise to the orientation of the drawings or to a culture plate in a normal position of use.

(13) Unless specified otherwise, the expressions “around”, “approximately”, “substantially” and “in the order of” signify within 10%, and preferably within 5%.

(14) The term transmission factor (or “transmittance”) of a layer designates the ratio of the intensity of the radiation coming out of the layer to the intensity of the radiation entering the layer. In the following description, a layer is said to be opaque to a radiation when the transmission factor of the radiation through the layer is smaller than 10%. In the following description, a layer is called transparent to a radiation when the transmission factor of the radiation through the layer is greater than 10%.

(15) The term reflection factor (or “reflectance”) of a layer designates the ratio of the light flow reflected by the layer to the light flow incident on the layer. In the following description, a layer is called reflective when the reflection factor of the layer is greater than 90%.

(16) FIG. 1 is a partial simplified perspective view of an embodiment of a multi-well culture plate or plate **100**.

(17) Culture plate **100** comprises an array of wells **102**. In the shown example, each well **102** has a cylindrical shape (of circular cross-section, in top view) vertically extending from the upper surface of plate **100**. Wells **102** are for example regularly distributed on the upper surface of plate **100**. As an example, all the wells **102** of plate **100** have identical dimensions, to within manufacturing dispersions.

(18) In the shown example, the wells **102** of plate **100** are recesses or compartments formed inside of a layer **104**. Wells **102** are for example intended to contain a biological substance (or medium) **106** for example comprising cells. As an example, biological substance **106** is dispensed into the wells **102** of plate **100** by means of a pipette **108** equipped with an adapted pipette tip adapted, for example, to the dimensions of wells **102**.

(19) Layer **104** is preferably made of a biocompatible material, in other words of a material which is non-toxic for the cells contained in biological substance **106**. As an example, layer **104** is made of polydimethylsiloxane (PDMS) or of a plastic polymer, for example, made of polystyrene, of polypropylene, of polycarbonate, etc. Layer **104** may be coated with a chemical functionalization intended to favor an adhesion of the cells contained in substance **106**. For example, the bottom and/or the lateral walls of at least a portion of wells **102** are functionalized.

(20) Culture plate **100** further comprises an array of light sources **112**. In the shown example, the array of light sources **112** is vertically aligned with respect to the array of wells **102**. Each light source **112** is more precisely located vertically in line with a well **102** to which it is associated. As an example, the bottom of each cell **102** is centered with respect to the upper surface of the associated light source **112**.

(21) In the shown example, the array of light sources **112** is coated on all its lateral surfaces with a layer **114**. As an example, layer **114** enables to protect against oxidation contacting elements (not shown) associated with light sources **112**. Layer **114** is for example a so-called underfill layer made of a biocompatible material. As an example, layer **114** is made of epoxy resin or glue, for example, the glue known under trade name "Epo-Tek MED-375".

(22) The light sources **112** of plate **100** are for example powered and controlled via a printed circuit board **116**. In the shown example, board **116** is located on the lower surface side of plate **100**, that is, the surface opposite to the array of wells **102**. Board **116** is for example coupled or connected to a control unit (not shown in FIG. 1), for example, via a ribbon cable **118** and a connector **120**. As an example, the control unit is a computer configured to supply the electric power and to control parameters of illumination of the light sources **112** of culture plate **100**.

(23) According to an embodiment, the light sources **112** of culture plate **100** are individually addressable. Each light source **112** may in particular emit or not light independently from the other light sources **112** of plate **100**. As an example, each light source **112** comprises a control circuit (not shown) which is specific thereto, that is, a control circuit independent from the control circuits of the other light sources **112** of plate **100**.

(24) In FIG. 1, the emission of light originating from a light source **112** has been symbolized by arrows starting from the upper surface of the considered light source **112** and oriented towards the inlet of the associated well **102**. In the shown example, it is arbitrarily assumed that four light sources **112** emit light while two light sources **112** emit no light.

(25) The illumination parameters of each light source **112** are preferably adjustable independently from the illumination parameters of the other light sources **112** of plate **100**. As an example, it is now possible to individually control, that is, for each light source **112** of plate **100**, parameters selected from among: an illumination duration; an emission wavelength or wavelength range, in other words an emitted color; a luminance or emission power level; and an emission mode, for example continuous or in pulsed light, it being for example possible to alternate the continuous and pulsed light emission modes during a same phase of exposure of a well **102**.

(26) Generally, culture plate **100** enables to control the illumination conditions within each well **102**. It may in particular be chosen to illuminate or not the biological substance **106** contained in each well **102** of plate **100** and to individually control the emission parameters of each light source **112** in operation.

(27) An advantage of culture plate **100** lies in the fact that it is possibly to vary, from one well **102** to the other, the illumination conditions to which biological substance **106** is submitted. In applications such as studies of cell response to light, optogenetics, neuromodulation, neuroprotection, diagnosis, and sequencing, the use of plate **100** for example enables to advantageously simplify the implementation of designs, in particular relating to the use of culture plates or plates having all their wells exhibiting similar illumination conditions.

(28) Another advantage of culture plate **100** lies in the fact that the materials of its external surfaces are for example compatible with methods of sterilization with ultraviolet light or with a gas, for example, ethylene oxide. As an example, this enables to reuse a same plate **100** to perform experiments implementing different biological substances.

(29) Although FIG. **1** shows an embodiment where culture plate **100** comprises six wells **102**, plate **100** may in practice comprise any number of wells **102**, for example, four, eight, twelve, twenty-four, forty-eight, or ninety-six wells **102**. Further, although FIG. **1** shows a culture plate **100** comprising wells **102** of cylindrical shape, plate **100** may in practice comprise wells **102** of any shape, for example, wells **102** having, in top view, a cross-section of rectangular or square shape.

(30) As an example, each well **102** has: a maximum dimension (in top view) in the order of from a few micrometers to a few millimeters, for example, in the range from 5 mm to 40 mm; and a depth in the range from 100 μm to 30 mm, for example, in the range from 10 mm to 20 mm.

(31) Each well **102** for example has: a surface (in top view) in the range from 0.19 cm.² to 13 cm.², for example, equal to approximately 0.3 cm.² for a plate **100** comprising ninety-six wells **102**; and a volume in the range from a few microliters to a few milliliters, for example in the range from 0.1 mL to 20 mL, for example, equal to approximately 0.3 mL for a plate **100** comprising ninety-six wells **102**.

(32) FIGS. **2** to **7** hereafter illustrate structures obtained at the end of successive steps of an example of an implementation mode of a method of manufacturing the multi-well culture plate **100** of FIG. **1**.

(33) FIG. **2** is a partial simplified cross-section view illustrating a step of the implementation mode of the method of manufacturing the culture plate **100** of FIG. **1**.

(34) During this step, power supply and addressing circuits **200** are formed inside and on top of a semiconductor substrate **202**.

(35) As an example, substrate **202** is a wafer or a piece of wafer made of a semiconductor material, for example, of silicon. In the orientation of FIG. **2**, power supply and addressing circuits **200** are formed on the upper surface or front surface side of substrate **202**. These circuits are in particular intended to enable to individually control each light source **112** (FIG. **1**) of plate **100**. As an example, each light source **112** is associated with a control circuit **200** distinct from the control circuits **200** of the other light sources **112** of plate **100**.

(36) FIG. **3** is a partial simplified cross-section view illustrating another step of the implementation mode of the method of manufacturing the culture plate **100** of FIG. **1**, formed from the structure such as described in relation with FIG. **2**.

(37) During this step, substrate **202** is thinned and vias **300** are formed.

(38) As an example, the thinning of substrate **202** is performed by grinding on the side of the surface of substrate **202** which does not comprise power supply and addressing circuits **200** (in other words, the lower surface of substrate **202**, in the orientation of FIG. **3**). At the end of this operation, substrate **202** for example has a thickness in the range from 50 μm to 150 μm , preferably from 70 μm to 100 μm .

(39) In the shown example, vias **300** cross substrate **202** across its entire thickness. Vias **300** for

example enable to transfer onto the lower surface of substrate **202** contacting elements of the circuits **200** located on the upper surface side of substrate **202**.

(40) The lateral walls and the bottom of through vias **300** are for example coated with portions **302** of an electrically-conductive layer, for example, a copper layer. In the shown example, each conductive layer portion **302** continues horizontally on top of and in contact with the lower surface of substrate **202**, in the vicinity of the corresponding via **300**.

(41) In practice, the conductive layer is for example deposited all over the structure on the lower surface side of substrate **202**. As an example, conductive layer portions **302** are then formed by etching.

(42) Substrate **202** further comprises, on its lower surface side, an electrically-insulating layer **304**. Layer **304** covers free areas of the lower surface of substrate **202**, and coats conductive layer portions **302** except for regions on top of and in contact with which connection pads **306** are formed.

(43) In practice, insulating layer **304** is for example first deposited all over the structure on the lower surface side of substrate **202**. As an example, openings are then formed in layer **304** to expose regions intended to receive connection pads **306**. Pads **306** are finally formed vertically in line with the regions which are not coated with insulating layer **304**.

(44) Although this is not shown, another insulating layer may be formed on the structure, on the lower surface side of substrate **202**, prior to the deposition of the conductive layer. In this case, this layer is for example intended to ensure a function of electric insulation between conductive layer portions **302** and substrate **202**.

(45) FIG. **4** is a partial simplified cross-section view illustrating still another step of the implementation mode of the method of manufacturing the culture plate **100** of FIG. **1**, carried out from the structure such as described in relation with FIG. **3**.

(46) During this step, light source array **112** is formed on the upper surface side of substrate **202**.

(47) According to an embodiment, each light source **112** comprises at least one organic light-emitting diode **400**, or OLED. In the shown example, each light source **112** comprises a single diode **400**. Each diode **400** is for example intended to emit white light.

(48) For simplification, each diode **400** is symbolized in FIG. **4** by a stack comprising an emissive layer **400E** interposed between a lower electrode **400B** and an upper electrode **400T**. In the shown example, the lower electrode **400B** of each diode **400** is on top of and in contact with the upper surface of substrate **202**.

(49) In practice, diodes **400** may have a more complex structure. As an example, each diode **400** may in particular comprise one or a plurality of additional layers (not shown) such as: an electron injection layer (EIL); an electron transport layer (ETL); a hole injection layer (HIL); and a hole transport layer (HTL).

(50) In the shown example, the lower and upper electrodes **400B** and **400T** of each organic light-emitting diode **400** respectively contact different portions of power supply and control circuits **200**. This for example enables to apply distinct electric potentials to lower electrode **400B** and to upper electrode **400T**. In the shown example, the lower and upper electrodes **400B** and **400T** of each diode **400** are connected to distinct pads **306** via different vias **300**.

(51) As an example, the lower electrode **400B** of each diode **400** reflects the light emitted by layer **400E** while upper electrode **400T** is transparent or semi-transparent. Layer **400E** for example has a transmittance greater than 10%. In the shown example, organic light-emitting diodes **400** are so-called top emission diodes.

(52) An implementation mode where each light source **112** comprises a single diode **400** emitting white light has been described in relation with FIG. **4**. As a variant, each light source **112** comprises a plurality of organic light-emitting diodes **400**, for example three organic light-emitting diodes **400** intended to respectively emit: red light, in other words an electromagnetic radiation having a wavelength for example in the range from 610 nm to 700 nm; green light, in other words an

electromagnetic radiation having a wavelength for example in the range from 500 nm to 570 nm; and blue light, in other words an electromagnetic radiation having a wavelength for example in the range from 450 nm to 490 nm.

(53) In the case where each light source **112** comprises a plurality of light-emitting diodes **400**, each diode **400** is preferably individually controllable. As an example, each diode **400** belonging to a same light source **112** is controllable independently from the other diodes **400** of the considered light source **112**, and independently from the diodes **400** forming part of the other light sources **112** of plate **100**.

(54) An advantage linked to the use of organic diodes lies in the fact that this enables to obtain a small spectral emission dispersion, for example in the order of from 10 nm to 100 nm. This dispersion for example results from the structure of organic diodes **400**. Biological substance **106** can thus be excited in limited wavelength ranges.

(55) FIG. 5 is a partial simplified cross-section view illustrating still another step of the implementation mode of the method of manufacturing the culture plate **100** of FIG. 1, carried out from the structure such as described in relation with FIG. 4.

(56) During this step, the upper surface of the structure is coated with a transparent planarization layer **500**.

(57) In the shown example, transparent planarization layer **500** particularly totally covers power supply and control circuits **200** as well as organic light-emitting diodes **400**. Layer **500** forms a hard coat.

(58) According to an embodiment, layer **500** is made of a photocrosslinkable organosilicon material. As an example, layer **500** is based on a material known under trade name HARDRISE of POLYRISE or a biocompatible resin. More generally, layer **500** is made of a biocompatible material. In practice, layer **500** is for example obtained by the implementation of a sol-gel-type method. Layer **500** for example has a thickness in the range from 1 μm to 10 μm .

(59) Although this is not shown, an encapsulation layer of organic light-emitting diodes **400** may be deposited on the upper surface side of substrate **202** prior to the deposition of layer **500**.

(60) FIG. 6 is a partial simplified cross-section view illustrating still another step of the implementation mode of the method of manufacturing the culture plate **100** of FIG. 1, carried out from the structure such as described in relation with FIG. 5.

(61) During this step, the array of wells **102** is formed.

(62) In practice, layer **104** is for example deposited over the entire surface of the structure. Wells **102** are then formed to expose portions of planarization layer **500** located vertically in line with each light source **112**. In the shown example, the bottom of each well **102** is formed by an upper surface portion of layer **500**.

(63) FIG. 7 is a partial simplified cross-section view illustrating still another step of the implementation mode of the method of manufacturing the culture plate **100** of FIG. 1, carried out from the structure such as described in relation with FIG. 6.

(64) During this step, the structure is linked to printed circuit board **116** and filling layer **114** is formed.

(65) Each pad **306** located on the lower surface side of substrate **202** is for example connected to a pad **700** located on the upper surface side of board **116**. As an example, an array of solder balls **702** enables to form the connection between pads **306** and pads **700**.

(66) In the shown example, layer **114** coats the lateral surfaces of layer **500**, of substrate **202**, and of layer **304**. Layer **114** further fills the free volume vertically delimited by the upper surface of board **116** and by the lower surface of layer **304**. Generally, layer **114** is deposited so as to protect pads **306**, solder balls **702**, and pads **700** against oxidation.

(67) An advantage of culture plate **100** lies in the fact that each light source **112** is separated from the bottom of well **102** to which it is associated by a distance shorter than 10 μm . Advantage is here taken from the integration of light sources **112** in culture plate **100** to decrease the distance between

the bottom of each well **102** and the associated diodes **400**. This for example enables to maximize the illumination to which the cells of the substance **106** present in well **102** are submitted, particularly as compared with a configuration where light sources of same power as sources **112** would be placed under a usual culture plate.

(68) Another advantage of culture plate **100** lies in the fact that organic light-emitting diodes **400** cause a lighter heat release than that which would be obtained by for example using inorganic light-emitting diodes. This for example enables to limit risks of degradation of the biological substance **106** contained in wells **102** under the effect of the heat released during the emission of light by diodes **400**.

(69) Another advantage of culture plate **400** lies in the fact that electric connections located on the upper surface of substrate **202** are transferred onto the lower surface of substrate **202** by means of through vias **300**. This enables, in particular, to protect circuits **200** against risks of aggressions which may for example result from the use of saline biological substances **106**. The presence, in plate **100**, of hard coat layer **500** particularly enables to avoid for biological substance **106** to come into contact with circuits **200** or diodes **400**. This for example provides a greater protection than a configuration where layer **500** would be opened to expose contacting elements located on the upper surface side of substrate **202**.

(70) Another advantage of culture plate **100** lies in the fact that its use is likely to allow a time gain, for example, in the context of sequencing methods comprising steps repeated in a loop, base by base, until an entire DNA (deoxyribonucleic acid) has been sequenced. As an example, the use of culture plate **100** may allow a time gain in a case where fluorophores, associated with complementary bases grafting on the DNA strand, are each excited by a different wavelength. In this case, three colors are for example sequentially “read” to determine the location of the first three bases, a lack of emission indicating the presence of the fourth base.

(71) Advantage may further be taken of culture plate **100** in DNA synthesis applications. As an example, the use of culture plate **100** may enable to do away with complex and time-consuming methods, such as methods comprising steps of protection and of deprotection by a microstructured mask (for example of resin type). The use of culture plate **100** in applications of this type may further enable to discard risks of errors linked, for example, to an exposure of biological substance **106** to ultraviolet light or to aggressive solvents.

(72) Various embodiments, implementation modes, and variants have been described. Those skilled in the art will understand that certain features of these various embodiments, implementation modes, and variants, may be combined and other variants will occur to those skilled in the art. In particular, it will be within the abilities of those skilled in the art to adapt the implementation mode of the method of manufacturing plate **100** described hereabove to a case where each light source **112** comprises a plurality of diodes.

(73) Finally, the practical implementation of the described embodiments, implementation modes, and variants is within the abilities of those skilled in the art based on the functional indications given hereabove. In particular, the forming of organic light-emitting diodes **400** is within the abilities of those skilled in the art based on the above indications.

Claims

1. Culture plate comprising: an array of light sources formed inside and on top of a semiconductor substrate; a transparent planarization layer totally coating the array of light sources and power supply and control circuits; and an array of wells formed on said layer, each light source in the array of light sources being located vertically in line with a well; wherein a bottom of each well in the array of wells is formed by an upper surface portion of the transparent planarization layer and the array of light sources is disposed below the bottom of the array of wells; wherein the transparent planarization layer includes a biocompatible material; wherein each light source

comprises at least one organic light-emitting diode; wherein each light source in the array of light sources comprises a respective independent control circuit; wherein the array of light sources is located on a side of a first surface of the substrate, opposite to a second surface of the substrate comprising connection pads; wherein connection pads are connected to the first surface of the substrate by through vias by an array of solder balls, to pads of a printed circuit board for powering and controlling the light sources.

2. Plate according to claim 1, wherein said layer is made of a photocrosslinkable organosilicon material.

3. Plate according to claim 1, wherein said layer has a thickness in a range from 1 μm to 10 μm .

4. Plate according to claim 1, wherein the array of wells is made of polydimethylsiloxane.

5. Plate according to claim 1, wherein each light source comprises a diode intended to emit white light.

6. Plate according to claim 1, wherein each diode is individually controllable.

7. Plate according to claim 1, wherein each light source comprises three diodes intended to respectively emit red light, green light, and blue light.

8. Plate according to claim 1, wherein bottom and/or lateral walls of at least a portion of the wells of the array are functionalized.

9. Method of manufacturing a plate according to claim 1.

10. Method according to claim 9, comprising the steps of: —forming the array of light sources inside and on top of the semiconductor substrate; —coating the array of light sources with the transparent planarization layer; and—forming at the surface of said layer a well vertically in line with each light source.

11. Plate according to claim 1, wherein the biocompatible material is polydimethylsiloxane (PDMS).

12. Plate according to claim 1, wherein the biocompatible material is plastic polymer.

13. Plate according to claim 12, wherein the plastic polymer is polypropylene, polystyrene, or polycarbonate.
